Epistemic Hygiene Arena-Uncued: A Diagnostic Pilot for LLM Agents in Polluted Evidence Environments

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May 22, 2026

Abstract

LLM agents increasingly reason over evidence environments that contain stale records, repost chains, generated pages, contradictory fragments, and source metadata that can become a shortcut. We introduce Epistemic Hygiene Arena-Uncued, a role-uncued diagnostic pilot for measuring whether a model reaches the right belief while keeping evidence roles clean and expressing useful verification actions. A previous cued construction exposed a benchmark-design failure: visible labels and document roles could leak the intended answer. That run is quarantined and is not used as evidence here. The new pilot removes direct material-role and conclusion cues from model-visible documents, checks leakage and shortcut baselines before model calls, and packages prompts, outputs, scorer artifacts, and readiness diagnostics. In a 60-task, four-model pilot with two neutral metadata views, all model calls parse successfully, but operational epistemic escape remains well below belief correctness. Generated-lore rows are the clearest failure mode: aggregate belief correctness is 0.938 while operational escape is 0.125 because models often keep polluted generated evidence in support fields. The release is diagnostic and pilot-scale. It is not a deployment certification, a universal leaderboard, or a claim that synthetic leakage has been fully eliminated.

1 Introduction

Information-seeking agents are often evaluated by final-answer accuracy. That is too coarse for settings where the evidence environment is polluted. A system can answer correctly while citing a repost, laundering generated text into a support field, or failing to specify the next verification action. These are not only presentation errors: downstream tools, auditors, and users may depend on the structured evidence and action fields.

Epistemic Hygiene Arena-Uncued tests this distinction in a small controlled setting. The pilot asks models to judge claims using synthetic evidence packets that contain clean evidence, conflicting evidence, false consensus, buried primary evidence, or generated lore. The model-visible packet does not label documents as primary, contaminant, generated, or gold-supporting. The scorer then separates belief correctness from evidence hygiene and active-verification behavior.

This paper replaces an earlier cued construction. That older run was useful as a failure discovery: it showed that visible construction cues could invalidate the benchmark. It is quarantined and not used as a main result. The contribution here is a role-uncued pilot with explicit pre-model leakage gates, shortcut baselines, four-model execution, scorer audit, and a reproducible artifact package.

The study is intentionally modest. It is a diagnostic pilot over 60 latent tasks, two neutral metadata views, four frontier model labels, and one clarified schema. It should be read as evidence that answer accuracy hides operational failures under controlled pollution, not as a stable ranking of model providers.

The main contributions are:

- a role-uncued version of the Epistemic Hygiene Arena task design that removes direct material-role and conclusion cues from model-visible evidence;
- pre-model leakage and shortcut-baseline gates showing that simple non-API shortcuts remain below the level needed to explain the strongest model results;
- a four-model pilot showing large gaps between belief correctness and operational epistemic escape, especially on generated-lore rows;
- a packaged artifact, verifier, and scorer audit that make the pilot reproducible and explicitly bounded.

2 Polluted Evidence Environments

Retrieval and tool-use benchmarks have shown that language models can use external evidence and actions, but the quality of the evidence environment itself is often treated as a background assumption [1, 4, 5]. In open-web settings, that assumption is fragile. Evidence may be stale, copied from another source, contradicted by a primary record, or generated in a form that looks plausible. Related work on factuality, retrieval-augmented generation, and poisoned retrieval motivates this concern [2, 3, 6].

Epistemic Hygiene Arena focuses on polluted evidence environments rather than only polluted final answers. The target behavior is epistemic escape: the model should form the right belief, select safe support, reject or avoid polluted evidence, and, when needed, express an executable verification action.

The role-uncued pilot is built around five evidence conditions:

- **Clean**: the packet contains enough clean evidence and no intended pollutant trap.
- **Conflicting evidence**: the packet contains evidence that pulls toward incompatible conclusions.
- **False consensus**: several surface-consistent records point the wrong way.
- **Buried primary**: the primary evidence is present but less salient than distractors.
- **Generated lore**: generated or lore-like material appears plausible but should not be used as clean support.

These conditions are synthetic by design. The point is not to mimic the full web, but to make failure modes measurable under controlled source relations.

3 EHA-Uncued Dataset Design

The pilot contains 60 latent tasks. Each task is rendered in two neutral metadata views, giving 120 model-visible task instances per model. The condition split is balanced: 12 clean, 12 conflicting-evidence, 12 false-consensus, 12 buried-primary, and 12 generated-lore tasks. The family split is 24 packet-judgment, 24 evidence-selection, and 12 active-verification tasks.

The key design change is role uncuedness. Model-visible documents use opaque per-task identifiers such as `doc_001`; visible titles, source types, timestamps, bodies, and citations do not expose

Table 1: Leakage and prompt-hygiene gate summary for the role-uncued pilot.

Gate	Rows/files	Hits	Status
Automatic leakage audit	480 docs	0 critical/high/medium	Passed
Hidden-label scan	480 docs	0 hidden-label hits	Passed
Semantic-ID/direct-cue scan	480 docs	0 semantic-ID or direct-answer hits	Passed
Local surface review	400 rows	0 critical leaks, 0 direct cues	Passed; not independent human validation
Prompt audit	480 prompts	0 hidden fields, 0 audit-ID hits	Passed
Stored hidden-label audit	480 outputs	0 prompt hits, 0 output hits	Passed

Table 2: Shortcut baseline summary. Values are operational epistemic escape.

Baseline family	Hidden view	Visible view	Interpretation
ID-only	0.000	0.000	Document IDs carry no useful signal
Cue phrase only	0.000	0.000	Direct cue phrases do not solve the task
Metadata/source/timestamp only	0.000	0.000	Neutral metadata alone is insufficient
Best non-heuristic shortcut	0.133	0.133	Below gate limit
Simple heuristic	0.350	0.333	Strongest shortcut; used for model comparison

construction roles such as primary support, contaminant, generated document, stale document, or gold verdict. Gold labels, dependency edges, and audit-only role labels are preserved for scoring but are not placed in model-visible prompts.

Two views test whether neutral metadata changes behavior without making roles explicit:

- `neutral_metadata_visible`: neutral metadata fields are visible.
- `neutral_metadata_hidden`: the same task is presented without those visible neutral metadata fields.

The pilot is not a replacement for a large benchmark. It is a validity-first dataset slice used to test whether the benchmark can survive basic leakage and shortcut checks before making model claims.

4 Leakage Controls And Shortcut Baselines

The older cued construction is quarantined because it exposed the wrong lesson: if benchmark construction roles are visible, models may solve the interface rather than the epistemic problem. The role-uncued pilot therefore runs leakage checks before model calls and records prompt-level audits during model execution.

Table 1 reports the main leakage controls. Automatic leakage checks found no critical, high, medium, hidden-label, semantic-ID, or direct-answer-cue hits. A local pre-model surface review covered 400 document rows and 50 unique tasks; it found no critical leaks or direct answer cues. This is not independent human-subject validation. It is a pre-model leakage gate used to decide whether model calls are justified.

Table 2 summarizes the no-API shortcuts. The strongest simple heuristic reaches 0.350 operational escape on the hidden view and 0.333 on the visible view. This is high enough to require caution, but low enough that model evaluation is not explained by document IDs, order, metadata alone, or a trivial cue phrase. The comparison is not uniformly favorable to every model-view pair: Claude and DeepSeek are at or below the simple heuristic on operational escape and evidence precision. The correct claim is narrower: the shortcut gates pass, and the strongest model-view pairs exceed the simple heuristic by meaningful margins.

5 Metrics And Scoring Contract

The scorer treats final-answer accuracy as one component, not the endpoint. The main metrics are:

- **Belief correctness**: whether the predicted verdict matches the gold verdict.
- **Evidence precision**: whether selected support avoids polluted evidence and aligns with clean support.
- **Clean support recall**: whether available clean support is recovered.
- **Polluted support rate**: how often polluted evidence is used as support.
- **Rejected pollutant rate**: how often known pollutants are rejected or avoided.
- **Uncertainty discipline**: whether confidence and insufficiency behavior follow the gold state.
- **Verification action score**: whether active-verification actions use the required action type and an executable target.
- **Operational epistemic escape**: the joint success criterion combining belief, evidence hygiene, and task-family-specific requirements.

Operational escape is intentionally stricter than answer accuracy. A model that answers correctly while placing generated lore in `supporting_evidence` should not receive full credit for an agent-facing evidence contract. Likewise, an active-verification answer can be correct while the next action is too vague to execute.

6 Experimental Setup

The pilot runs four model labels: `gpt-5.5`, `claude-opus-4-7`, `gemini-3.1-pro-preview`, and `deepseek-v4-pro`. Each model receives 120 task instances: 60 latent tasks times two neutral metadata views. The schema variant is `clarified`; the prompt condition is `standard_answer`.

All 480 model records parse successfully. The stored prompt audit reports zero semantic document-ID hits, zero semantic visible-citation hits, zero audit-ID title/body hits, and zero hidden-field hits across 480 prompts. The stored hidden-label audit reports zero prompt hits and zero output hits. The total Phase 12 model-call cost recorded by the runbook is USD 3.764007.

DeepSeek required special handling before the final run. A DeepSeek-only preflight retry used `response_format=json_object`, first-JSON-object extraction, omitted temperature, `max_completion_tokens=4096`, merged developer/system content into the user message, disabled LLM repair, and recorded the invocation profile separately. The Phase 12 run uses the same documented profile with a maximum of two attempts.

Table 3: Main model metrics by neutral metadata view. Operational escape is the joint success metric.

Model	Hidden op.	Hidden belief	Hidden evidence	Visible op.	Visible belief	Visible evidence
`gpt-5.5`	0.517	1.000	0.758	0.533	0.983	0.750
`gemini-3.1-pro-preview`	0.417	0.950	0.877	0.450	0.883	0.808
`claude-opus-4-7`	0.300	0.933	0.500	0.333	0.933	0.463
`deepseek-v4-pro`	0.217	0.667	0.435	0.250	0.650	0.447

Table 4: Aggregate model metrics by evidence condition across all four models and both views.

Condition	n	Operational	Belief	Evidence precision	Polluted support
Clean	96	0.719	0.927	0.994	0.000
Buried primary	96	0.490	0.979	1.000	0.000
False consensus	96	0.302	0.719	0.659	0.323
Conflicting evidence	96	0.250	0.812	0.479	0.448
Generated lore	96	0.125	0.938	0.000	0.802

7 Results

Table 3 gives the main model metrics by neutral metadata view. Operational escape is lower than belief correctness for every model and view. `gpt-5.5` has the highest operational escape in both views, at 0.517 hidden and 0.533 visible. `gemini-3.1-pro-preview` follows at 0.417 hidden and 0.450 visible. Claude and DeepSeek have lower operational escape despite high parse success.

Table 4 shows that the aggregate difficulty is not uniform. Clean rows are much easier, with 0.719 operational escape. Generated-lore rows have 0.938 belief correctness but only 0.125 operational escape because evidence precision collapses to 0.000 and polluted support remains high.

Table 5 makes the separation explicit. The largest belief-to-operation gap appears in generated lore. Buried-primary rows also separate: belief correctness is 0.979, while operational escape is 0.490. This supports the core claim that the task is not just asking whether the model can answer the claim. It asks whether the model can keep the evidence contract clean enough for an agent pipeline.

Table 6 summarizes active-verification action behavior. Exact target rate is 1.000 in this pilot, so the remaining variation comes from required action recall and the action score. This result should be read cautiously because the active-verification slice has only 96 rows.

8 Schema/Interface Analysis

The current artifact uses the clarified schema for all main results. The earlier cued construction motivated a schema-sensitivity question, but its model results are quarantined and are not used here. Phase 15 therefore records the schema-ablation mini-slice as planned Phase 1.1 work, not as a completed result.

The interface finding that remains valid in this pilot is narrower. Structured fields matter because evidence and action fields can fail while the final verdict succeeds. In generated-lore rows, models often know the right verdict yet fail operational escape because polluted evidence remains in support fields. In active-verification rows, the action schema makes executable next steps measurable instead of leaving them as prose.

Future schema work should run a small role-uncued ablation over generated-lore and buried-primary tasks using the planned variants `clarified`, `current`, `minimal`, and `diagnostic_no_hygiene`. Until that ablation is run, this paper does not claim that schema wording causes the observed gaps.

Table 5: Belief correctness versus operational escape by condition.

Condition	Belief	Operational	Gap	Evidence precision
Generated lore	0.938	0.125	0.812	0.000
Buried primary	0.979	0.490	0.490	1.000
False consensus	0.719	0.302	0.417	0.659
Conflicting evidence	0.812	0.250	0.562	0.479
Clean	0.927	0.719	0.208	0.994

Table 6: Active-verification action exactness by model. Each model contributes 24 active-verification rows.

Model	Action score	Exact target	Required recall	Overall operational
`claude-opus-4-7`	0.875	1.000	0.875	0.317
`gemini-3.1-pro-preview`	0.812	1.000	0.812	0.433
`gpt-5.5`	0.792	1.000	0.792	0.525
`deepseek-v4-pro`	0.688	1.000	0.688	0.233

9 Human Scorer Audit

Table 7 reports the Phase 14 scorer audit. The audit reviewed 40 rows balanced across models, views, conditions, task families, and operational successes/failures. It found no scorer disagreements, no systematic scorer bug, and no need for rescoring.

The audit is still limited. It is a local Codex-assisted manual scorer audit, not an independent human-subject review. The claim is therefore restricted to scorer-contract sanity for this pilot, not external validation of every action-quality judgment.

10 Limitations

The pilot is small: 60 latent tasks, two neutral metadata views, four model labels, and one prompt/schema condition. It is not statistically powered for fine-grained provider ranking. Reported model differences should be treated as descriptive.

The tasks are synthetic. Synthetic control is useful for isolating source-role failures, but it may introduce prose regularities that real evidence environments do not share. The leakage and baseline gates reduce this risk; they do not eliminate it.

The human review evidence has two different scopes. The 400-row pre-model surface review is a local leakage gate. The 40-row scorer audit is a local Codex-assisted manual audit. Neither should be described as independent human validation.

The schema-ablation mini-slice was skipped under the runbook rule and deferred to Phase 1.1. This paper can motivate schema/interface questions, but it cannot claim that a role-uncued schema ablation has already been run.

Finally, the artifact includes hidden labels and gold data for reproducibility, but those files are not model-visible. Any downstream reuse must preserve that separation.

Table 7: Phase 14 scorer audit summary.

Audit item	Value
Reviewed rows	40
Models covered	4, 10 rows each
Views covered	20 hidden, 20 visible
Conditions covered	8 rows each
Families covered	16 packet, 16 evidence, 8 active
Operational successes/failures	20 / 20
Scorer disagreements	0
Systematic scorer bug found	false
Rescoring required	false
Independent human review	false

11 Ethics And Scope

Epistemic Hygiene Arena-Uncued is a diagnostic benchmark for controlled polluted evidence environments. It is not a safety certification, a deployment approval test, or a universal leaderboard. High performance on this pilot would not establish that an agent is reliable in real legal, medical, scientific, or financial workflows.

The benchmark uses synthetic entities and synthetic evidence packets. This avoids collecting sensitive real-user data and makes gold provenance explicit. The tradeoff is ecological validity: synthetic documents cannot cover the full distribution of open-web evidence, institutional records, or adversarial manipulation.

The intended use is research analysis of failure modes: belief formation, evidence-role hygiene, shortcut sensitivity, and verification-action expression. The intended non-use is automated certification of deployed systems without additional task coverage, independent audits, domain-specific validation, and human oversight.

12 Artifact

The release-facing artifact is rooted at `artifact_uncued_phase1/`. It includes the role-uncued pilot data, prompt artifacts, schemas, scorer files, model outputs, baselines, audits, examples, a minimal reproduction script, and an artifact verifier script. The manifest records 1022 hashed files and explicitly marks `cued_artifacts_included=false`.

The main verifier is:

```
uvruneha-verify-uncued-phase1
--artifact-dir../artifact_uncued_phase1
--reports-dir../reports
--out-dir../reports
```

The Phase 17 readiness report passes all artifact gates: layout, manifest hashes, visible-file hidden-label scan, gold/scorer inclusion, prompt inclusion, leakage gate, baseline gate, human leakage gate, model-run completeness, scored-result completeness, scorer audit, and explicit Phase 15 schema-ablation status. The minimal no-dependency smoke check is `artifact_uncued_phase1/reproduce_minimal.sh`.

The older `artifact/` package is quarantined as cued/internal evidence and is not a source for the main results in this paper.

13 Conclusion

Epistemic Hygiene Arena-Uncued shows that, in a controlled role-uncued pilot, answer accuracy can substantially overstate agent readiness in polluted evidence environments. The clearest evidence is generated lore: models usually reach the right belief, yet operational escape remains low because evidence roles are not clean. Active-verification rows show a second interface requirement: a useful agent must express the next verification action in a machine-checkable form.

The release should be read as a validity-first pilot. It fixes the direct cueing failure exposed by the older run, packages the uncued artifacts, and records leakage, baseline, model-run, scorer-audit, and readiness diagnostics. The next step is not leaderboard expansion alone; it is a larger role-uncued task set with independent human audit and a planned schema ablation.

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